



RDNA2 Micro-Architectural Optimization for MoE PCIe Latency Hiding: A Comprehensive Research Report



Executive Summary

The performance ceiling for massive Mixture-of-Experts (MoE) models on the AMD Radeon RX 6800 XT is not defined by its computational throughput, but by its ability to sustain data flow across the PCIe 4.0 x16 interconnect [[1]]. With models like `Qwen3_35BMTPIQ4.gguf` exceeding the GPU's 16 GB of VRAM, a significant portion of the model must reside in system RAM and be dynamically streamed into the GPU during inference [[1]]. This creates a profound architectural mismatch: a high-bandwidth compute engine is starved of data by a comparatively narrow I/O channel. The bottleneck has shifted from VALU utilization to the hardware latency cascade inherent in Host-to-Device (H2D) transfers, kernel synchronization overhead, and inefficient memory access patterns [[1]].

This report presents a comprehensive, ISA-driven optimization strategy designed specifically to eliminate this latency cascade. It is grounded not in speculative software engineering, but in the concrete, low-level instructions and micro-architectural features documented in the *RDNA2 Shader Instruction Set Architecture* manual [[1]]. The analysis confirms that the most impactful optimizations are immediate, low-level techniques focused on three interconnected pillars: **asynchronous SDMA pipelining**, **zero-copy GTT access with precise cache control**, and **ultra-low-overhead wavefront synchronization** [[1]]. These are not abstract concepts; they are direct mappings to specific hardware primitives such as `s_sendmsg`, `GLOBAL_LOAD_DWORD` with `SLC=1`, and `s_sem_wait` [[1]].

The strategy is predicated on two non-negotiable boundary conditions. First, the macro-level MoE offloading strategy—splitting experts between VRAM and CPU—is fixed. The goal is not to redesign the algorithm, but to ensure the hardware-level execution pipelines running that algorithm contain zero assembly-level execution bubbles [[1]]. Second, pinned host memory (GTT) is an absolute prerequisite. Attempting to optimize

for unpinned memory would be futile, as the resulting OS page-fault latencies would completely dwarf any micro-optimization gains [[1]]. Therefore, all recommendations assume the use of `hipHostMalloc` or `hipHostRegister` for all host-side allocations [[1]].

This report provides a complete end-to-end blueprint, moving from the theoretical foundations of the RDNA2 ISA to the concrete implementation details required for integration into the `ggml-hip` backend. It details the exact assembly mechanisms, the precise location within the codebase where each modification should be made, and the critical synchronization contracts that must be established between the GPU kernels and the CPU orchestrator. The result is a path to transform the problem from one of raw bandwidth limitation into one of efficient, hardware-accelerated orchestration—a problem that the RDNA2 silicon was explicitly engineered to solve [[1]].



Foundational Context: The MoE PCIe Bottleneck on RDNA2

To understand the necessity of the proposed micro-architectural optimizations, one must first grasp the nature of the fundamental constraint. The RX 6800 XT is built around the RDNA2 architecture, which features a sophisticated memory hierarchy designed for high-throughput graphics workloads [[1]]. Its primary components include a large, on-die Infinity Cache (L3), per-CU Local Data Share (LDS) memory, and a robust connection to external GDDR6 memory via a high-speed bus [[1]]. However, when confronted with MoE models that exceed the 16 GB VRAM capacity, the GPU is forced to rely on system RAM, accessed over the PCIe 4.0 x16 interface [[1]]. While this interface offers a theoretical bandwidth of ~ 31.5 GB/s, it introduces a new class of latency that is orders of magnitude higher than accessing local VRAM or even the Infinity Cache [[1]].

The performance degradation is not linear. It manifests as a severe "latency cascade" [[1]]. When the MoE routing logic determines that a particular expert layer is required, the CPU orchestrator must initiate a transfer. This process involves several sequential, blocking steps: the CPU must prepare the memory region, issue a `hipMemcpyAsync` command, wait for the command to be enqueued, and then wait for the transfer to complete before signaling the GPU that the weights are ready for computation. During this entire sequence, the GPU's powerful VALU units sit idle, waiting for data. This is the root cause of the observed -7.3% decode regression in the current `main` branch compared to the `v0.3.0-stable` baseline [[1]]. The compile-time definitions were

poisoning the optimizer, but the underlying reason for the fragility is this latent I/O bottleneck: the system is so sensitive to any inefficiency in the data pipeline that even minor changes in register allocation can disrupt the delicate balance needed to hide PCIe latency [[1]].

The telemetry evidence is unambiguous. Profiling the `mul_mat_vec_q` kernel—the hot path for inference—reveals that the dominant stall reason is `WAVE_ISSUE_WAIT`, accounting for 52,560 cycles, or 65% of the total dispatch time [[1]]. This is not a sign of insufficient computational power; it is a clear signal that the scheduler is starved of independent instructions because the entire execution pipeline is synchronized to the pace of the slowest link: the PCIe bus [[1]]. Furthermore, the `MemUnitBusy` metric stands at 85.29%, indicating that the memory unit is saturated, but not fully utilized [[1]]. This gap suggests that the bottleneck is not raw bandwidth, but rather the *efficiency* with which that bandwidth is being used. The memory unit is spending a significant amount of time waiting for commands to be issued, waiting for coherency protocols to resolve, or waiting for dependencies to clear, rather than actively transferring data. This points directly to suboptimal memory access patterns and poor overlap between transfer and computation.

The standard software solutions are inadequate for this level of optimization. High-level abstractions like `hipMemcpyAsync` introduce their own layers of overhead and cannot guarantee the precise timing and instruction-level control required to achieve perfect overlap. They are designed for generality, not for squeezing the last nanosecond of performance out of a specific hardware configuration. Similarly, generic asynchronous stream management does not address the core issue: the GPU is still fundamentally waiting for a signal from the CPU, creating a hard dependency. To break this cycle, we must move down the stack, past the driver and runtime APIs, and into the realm of the shader ISA itself. We must leverage the hardware's native capabilities for autonomous DMA operations, direct memory access, and ultra-low-latency inter-process communication. The RDNA2 ISA manual is not a document of abstract theory; it is a detailed map of the physical pathways and control registers that govern the flow of data and instructions across the entire SoC [[1]]. Our task is to read this map and build an optimized route that bypasses every known bottleneck.



ISA-Based Optimization Blueprint: Pillar 1 — Advanced SDMA Pipelining & H2D Overlapping

The first pillar of our optimization strategy targets the most visible symptom of the PCIe bottleneck: the GPU sitting idle while waiting for data. The solution is not to make the transfer faster, but to make the GPU do useful work *while* the transfer is happening. This is the principle of overlapping execution, and RDNA2 provides dedicated hardware to achieve it at the lowest possible level: the Asynchronous Memory Engines (ACEs) and the System DMA (SDMA) units [[1]].

Hardware-Level Instruction Synchronization: Autonomous SDMA Triggering

The traditional approach to overlapping is to launch a `hipMemcpyAsync` from the CPU, followed by launching a GPU kernel. The kernel might then need to synchronize with the completion of the copy using `hipStreamSynchronize`. This creates a software-defined loop with multiple round-trip latencies. The RDNA2 ISA offers a far more elegant solution: allowing the GPU itself to trigger the next transfer autonomously [[1]]. This is achieved through the `s_sendmsg` instruction, which is used to send messages to various hardware blocks on the chip. By targeting the SDMA queue with a specific message, a shader kernel can initiate a DMA operation without involving the CPU at all [[1]].

The key insight is that the `s_sendmsg` instruction can be configured with the `MSG_GDS` type, which allows it to write directly to the Global Data Share (GDS) memory space [[1]]. In practice, this means the kernel can write a descriptor containing the source address (in pinned host memory), destination address (in VRAM), and size directly into a pre-allocated SDMA descriptor ring buffer. Once written, the SDMA hardware automatically reads the descriptor and executes the transfer. This transforms the data movement from a synchronous, CPU-controlled process into an asynchronous, GPU-autonomous pipeline. The `mul_mat_vec_q` kernel, after finishing its computation on the current expert's weights, can immediately issue the `s_sendmsg` call to load the next expert's weights, ensuring that the VRAM buffer is populated before the next iteration begins [[1]].

The `s_sendmsg` instruction is highly configurable. Its operand includes a bitfield that specifies the target engine and the message type. For SDMA, the relevant bits are set to `0x2` for the `MSG_SDMA` message type [[1]]. The instruction also takes a source register (`s_rc`) that contains the 32-bit value to be sent. This value is typically a pointer to the descriptor structure. The descriptor itself is a small, fixed-size structure that must be laid

out in memory according to the SDMA specification. It will contain fields such as `SRC_ADDR_LO`, `SRC_ADDR_HI`, `DST_ADDR_LO`, `DST_ADDR_HI`, `BYTE_COUNT`, and `CONTROL`. The exact format and bit layout for these fields are specified in the RDNA2 ISA manual, particularly in the chapters covering the Asynchronous Memory Engines and SDMA [[1]]. The critical point is that the kernel can construct this descriptor in registers or LDS and then write it to the SDMA queue with a single, low-overhead instruction.

Optimal Memory Fence Usage: Ensuring Safe Overlap

Autonomous SDMA triggering is only half the battle. The other half is ensuring that the data loaded by the SDMA engine is safely available for consumption by subsequent instructions. If the GPU attempts to read from the VRAM buffer before the SDMA transfer is complete, it will encounter stale data or undefined behavior. The RDNA2 ISA provides a precise, hardware-synchronized mechanism for this: the combination of `s_dcache_wb` and `s_waitcnt` [[1]].

The `s_dcache_wb` instruction is a scalar ALU (SALU) instruction that writes back any dirty data from the L1 data cache to the L2 cache (or to memory, depending on the cache policy) [[1]]. Its primary purpose here is to act as a memory barrier, ensuring that all previous memory writes—such as those performed by the kernel for updating the KV cache or writing intermediate results—are completed and visible before proceeding. This prevents a scenario where the SDMA transfer overwrites data that is still pending in the cache.

Following `s_dcache_wb`, the `s_waitcnt` instruction is used to stall the SALU until a specific condition is met [[1]]. The most critical parameter for our use case is `vmcnt`, which stands for "vector memory count." This counter tracks the number of outstanding vector memory operations (VMEM) that have been issued but not yet completed [[1]]. By executing `s_waitcnt(vmcnt(0))`, the SALU will stall until the `vmcnt` reaches zero, meaning that *all* previously issued VMEM loads and stores have finished. This is the definitive hardware signal that the SDMA-initiated transfer has successfully written the new expert weights into the designated VRAM buffer and that the data is now safe to read [[1]].

The sequence is therefore: 1. Execute `s_dcache_wb` to flush any pending writes. 2. Issue the `s_sendmsg` to start the SDMA transfer. 3. Execute `s_waitcnt(vmcnt(0))` to stall until the SDMA transfer completes.

This sequence is not a guess; it is a direct mapping to the hardware's guaranteed behavior. The `s_waitcnt` instruction is a fundamental part of the RDNA2 execution

model, and its interaction with the `vmcnt` counter is a well-defined, deterministic hardware feature [[1]]. This eliminates the need for polling loops or complex event-based synchronization, both of which consume valuable compute resources and introduce unpredictable latency.

The benefit of this approach is twofold. First, it guarantees correctness: the kernel will never read stale data. Second, it maximizes efficiency: the `s_waitcnt` is a very low-overhead stall. Unlike a busy-wait loop, which consumes power and occupies the SALU, the `s_waitcnt` puts the SALU into a low-power state while it waits, freeing up the CU for other waves to execute. This is the essence of RDNA2's "never-idle" design philosophy [[1]].

Implementation Blueprint: SDMA Integration in `ggml-hip`

The implementation of this SDMA pipelining strategy requires modifications to the core matrix-vector multiplication kernel, as this is the central hot path where the data transfer and computation are most tightly coupled. The logical location for this change is within the `mul_mat_vec_q` function in `ggml/src/ggml-hip/mmvq.cu` [[1]].

The modification is surgical. After the main computational loop has processed the current block of weights, a new code segment should be inserted to handle the prefetching of the next block. This segment will be guarded by a conditional check to ensure it only runs when there is a next block to load. The pseudo-code for this section is:

```
// Pseudo-code for SDMA prefetch in mul_mat_vec_q
if (kbx + 1 < nb_per_thread) {
    // Step 1: Construct the SDMA descriptor in registers
    uint32_t src_addr_lo = get_host_address_for_block(kbx + 1);
    uint32_t src_addr_hi = get_host_address_high(src_addr_lo);
    uint32_t dst_addr_lo = get_vram_buffer_address();
    uint32_t dst_addr_hi = get_vram_address_high(dst_addr_lo);
    uint32_t byte_count = get_weight_block_size();

    // Step 2: Write descriptor to SDMA queue (s_sendmsg)
    // This assumes 'descriptor' is a register containing the descriptor
    __builtin_amdgcn_s_sendmsg(__builtin_amdgcn_s_sendmsg(MSG_SDMA, descri

    // Step 3: Ensure prior writes are complete and wait for SDMA
    __builtin_amdgcn_s_dcache_wb();
```

```
    __builtin_amdgcn_s_waitcnt(0x100); // vmcnt(0)
}
```

The actual implementation will require defining the SDMA descriptor structure and populating it correctly. This structure is not a C++ `struct` but a sequence of 32-bit values laid out in memory. The `s_sendmsg` instruction will write the base address of this descriptor to the SDMA queue's doorbell register, instructing the hardware to begin processing it [[1]].

The `get_host_address_for_block()` function would be responsible for calculating the address of the next expert's weight block in pinned host memory. This calculation must be based on the MoE routing decision, which is determined by the CPU orchestrator. Therefore, the kernel must be provided with a pointer to a lookup table or a pre-computed address, likely passed in as a kernel argument or stored in a constant memory region.

The `get_vram_buffer_address()` function would return the address of a pre-allocated, circular buffer in VRAM. This buffer serves as the staging area for incoming weight data. Using a circular buffer allows the kernel to manage multiple in-flight transfers, preventing a situation where the SDMA engine is blocked waiting for a buffer to become free.

This implementation achieves the core objective: it decouples the CPU from the transfer initiation process. The CPU only needs to perform the initial setup and pass the necessary pointers to the kernel. Once launched, the GPU kernel manages its own data pipeline, initiating the next transfer as soon as it knows it will need it. This eliminates the CPU-GPU handshake latency and creates a true, self-sustaining data pipeline that can keep the VALU units fed continuously.



ISA-Based Optimization Blueprint: Pillar 2 — Host-Access Memory (Zero-Copy GTT) & Cache Control Flags

While SDMA pipelining addresses the latency of moving large chunks of expert weights, the MoE inference workflow also involves frequent, smaller accesses to gating networks and routing logic. These accesses are often to small matrices that determine which expert should be activated for a given token. Performing a full `hipMemcpyAsync` for each of these small lookups would be catastrophically inefficient, introducing a massive number

of small, high-latency transactions. The second pillar of our strategy leverages RDNA2's advanced memory architecture to allow the GPU to access these small, frequently-used data structures directly from pinned system RAM, a technique known as Zero-Copy Graphics Translation Table (GTT) access [[1]].

Cache Coherency Policy Tuning: Bypassing the Cache Hierarchy

The key to effective GTT access lies in understanding and controlling the memory coherency policies. The RDNA2 ISA defines several flags that modify how a memory load instruction interacts with the cache hierarchy: GLC (Global Level Coherence), SLC (System Level Coherence), and DLC (Device Level Coherence) [[1]]. Each flag represents a different trade-off between latency, bandwidth, and cache pollution.

For MoE workloads, the optimal strategy is to *avoid* caching the expert weight data altogether. The reason is simple: the L1 and L2 caches are extremely valuable resources, primarily dedicated to storing the "hot" KV cache data, which is accessed repeatedly for every token generation step [[1]]. If the "cold" expert weight data is allowed to flood these caches, it will evict the hot KV data, causing a catastrophic increase in cache misses and forcing the GPU to fetch the same KV data repeatedly from slower VRAM or even the Infinity Cache. This cache thrashing is a major contributor to the observed MemUnitBusy saturation at 85.29% [[1]].

Therefore, the correct configuration is to set both the GLC and SLC flags to 1. Setting GLC=1 ensures that the load is coherent with all agents on the chip, including the CPU, which is essential for maintaining consistency between the CPU's view of the data and the GPU's view [[1]]. Setting SLC=1 is the crucial step: it forces the load to go directly to the system's memory controller, bypassing the L1 and L2 caches entirely [[1]]. The data flows straight from the DRAM over the PCIe bus and into the GPU's memory subsystem, but crucially, it never touches the precious cache lines reserved for the KV cache. This is not a compromise; it is an optimization that preserves the integrity of the fast path for the most critical data.

Direct VGPR Loading Strategy: Streaming Data into the Compute Pipeline

Having decided to bypass the cache, the next question is: where should the data go? The answer is the most direct path to the computational units: the Vector General-Purpose Registers (VGPRs). Loading data directly into the VGPR file eliminates the need for an additional, costly step of reading from VRAM into a temporary register. It creates a direct streaming pipeline from the PCIe bus into the heart of the compute engine.

The RDNA2 ISA provides the `GLOBAL_LOAD_DWORD` instruction for precisely this purpose [[1]]. This instruction performs a 32-bit load from a global memory address. To achieve the desired zero-copy, direct-to-VGPR behavior, it must be used in conjunction with the `SLC=1` flag. The instruction's encoding includes a field for specifying the cache policy, and setting the appropriate bits for `SLC=1` configures it to perform a direct, uncached load [[1]].

The assembly sequence for loading a single gating network weight would look something like this:

```
global_load_dword v0, s[0:1], 0x00000000 slc
```

Where `v0` is the destination VGPR, `s[0:1]` contains the 64-bit address of the data in pinned host memory, and `slc` is the modifier that sets the `SLC=1` flag [[1]].

This instruction is exceptionally efficient. It issues a single, atomic request to the memory controller and places the result directly into the requested VGPR. There is no intermediate storage in VRAM, no cache line fill, and no subsequent `load` instruction required. The data is available for the next `v_fma_f32` instruction in the following cycle. This is the most direct and lowest-latency path possible for accessing host memory on the RDNA2 architecture [[1]].

Implementation Blueprint: Zero-Copy GTT Access in `ggml-hip`

The implementation of zero-copy GTT access should be integrated into the parts of the codebase responsible for fetching small, dynamic data structures that drive the MoE routing logic. The most appropriate location is `ggml/src/ggml-hip/ggml-hip-get-rows.cuh`, which contains functions for retrieving rows of data from matrices [[1]]. This file already deals with the abstraction of "getting rows," making it the natural place to introduce a specialized, optimized path for accessing gating network weights.

The existing `ggml_get_rows_hip` function could be modified to include a new, optimized path for gating network lookups. This path would be triggered by a specific flag or context, indicating that the requested row is part of a MoE gating matrix.

The core of the new implementation would be a custom inline assembly block that replaces the standard `hipMemcpyAsync` call. This block would: 1. Load the address of the requested gating network weight from a pointer passed to the kernel. 2. Use the `GLOBAL_LOAD_DWORD` instruction with the `slc` modifier to perform the direct, uncached load into a VGPR. 3. Perform the necessary arithmetic (e.g., scaling, bias addition) on the loaded data using the VALU.

The C++ wrapper would look like this:

```
// In ggml-hip-get-rows.cuh
__device__ inline float get_gating_weight_direct(const float* __restrict__
    float weight;
    asm volatile (
        "global_load_dword %0, [%1], 0x00000000 slc"
        : "=v"(weight)
        : "s"(gating_weights + idx)
        : "memory"
    );
    return weight;
}
```

The `asm volatile` block is a direct translation of the ISA instruction into inline assembly. The `"=v"(weight)` output constraint tells the compiler to store the result in a vector register, and the `: "s"(gating_weights + idx)` input constraint tells it to load the address into a scalar register pair (`s[0:1]`). The `"memory"` clobber informs the compiler that this assembly block may read from memory, preventing unwanted optimizations that could reorder the load.

This implementation is not a replacement for all memory transfers; it is a targeted, surgical tool for the most latency-sensitive, small-data-access patterns. By using this method for gating network lookups, the GPU can make its MoE routing decisions with near-zero latency, eliminating a major source of pipeline stalls and enabling the VALU units to remain active and productive throughout the entire inference sequence [[1]].

ISA-Based Optimization Blueprint: Pillar 3 — Asynchronous Multi-Stream Synchronization & Signaling

The final pillar addresses the most subtle yet critical bottleneck: the handoff between the CPU orchestrator and the GPU worker. Even with perfect SDMA pipelining and zero-copy GTT access, the system can stall if the GPU is forced to wait for the CPU to signal that a new expert is ready. Traditional synchronization methods like `hipStreamSynchronize` are expensive, involving a round-trip to the driver and a context switch. This overhead is unacceptable for the tight, high-frequency loops of MoE

inference. RDNA2 provides dedicated hardware primitives for this exact purpose: ultra-low-overhead, hardware-assisted signaling and stalling.

Ultra-Low-Power Wavefront Stalling: Retaining CU Occupancy

The RDNA2 ISA provides the `s_sleep` instruction, which is designed to put a wavefront into a deep, low-power sleep state [[1]]. This is fundamentally different from a software busy-wait loop. A busy-wait loop keeps the SALU active, consuming power and preventing other waves from utilizing the CU. In contrast, `s_sleep` relinquishes the SALU while keeping the wavefront's state intact in the CU's wavefront scheduler [[1]]. This allows other, independent wavefronts to execute on the same CU, maximizing hardware utilization. When the wake-up condition is met, the sleeping wavefront resumes instantly, with no loss of context.

The `s_sleep` instruction is configured with a mask that specifies which memory locations to monitor [[1]]. The CPU orchestrator can write to a specific, shared memory address, and the GPU can configure `s_sleep` to watch for a change at that address. This creates a direct, lock-free communication channel. The GPU kernel, upon completing the processing of the current expert and issuing the SDMA transfer for the next, can simply execute `s_sleep` and wait for the CPU to update the shared semaphore. This is a passive, zero-power operation that perfectly hides the latency of the CPU's expert selection and preparation process [[1]].

Hardware User-Signal Integration: Lock-Free Communication

An alternative, and potentially more efficient, primitive is the `s_sem_wait` instruction [[1]]. This instruction is specifically designed for semaphore-style synchronization. It allows the GPU to wait for a specific integer value to reach a certain threshold, typically by waiting for a counter to be incremented by the CPU [[1]]. Like `s_sleep`, it is a hardware-accelerated, low-overhead operation that avoids the pitfalls of software-based polling.

The `s_sem_wait` instruction works by taking a memory address and a comparison value as operands [[1]]. The GPU will stall until the value at the specified address is greater than or equal to the comparison value. This is ideal for implementing a producer-consumer pattern between the CPU and GPU. The CPU acts as the producer, incrementing a semaphore in shared memory each time a new expert is prepared and ready for loading. The GPU acts as the consumer, waiting for that semaphore to indicate readiness.

The advantage of `s_sem_wait` over `s_sleep` is its precision. `s_sleep` is a general-purpose stall, while `s_sem_wait` is a purpose-built, atomic operation for inter-process signaling. It is less susceptible to race conditions and provides a cleaner, more semantically clear contract between the two processors [[1]].

Implementation Blueprint: Wavefront Synchronization in `ggml-hip`

The implementation of hardware signaling must be distributed across both the GPU kernel and the CPU orchestrator. The kernel modification is relatively straightforward, while the CPU side requires establishing a new, explicit API contract.

In `ggml/src/ggml-hip/convert.cu`, the expert weight loading kernel should be modified to include the synchronization primitive at the beginning of its execution. The kernel should be designed to be launched in anticipation of the next expert, but to stall until the CPU signals it is ready. The pseudo-code for this is:

```
// In convert.cu, inside the expert weight loader kernel
extern __constant__ uint32_t g_expert_semaphore;

// Wait for the CPU to signal that the expert is ready
__builtin_amdgcn_s_sem_wait(&g_expert_semaphore, 1);

// Now proceed with the weight loading logic...
```

The `g_expert_semaphore` is a `__constant__` variable, which is mapped to a fixed, known address in the GPU's memory space. The CPU orchestrator will write to this same address to signal readiness [[1]].

On the CPU side, the main `ggml` orchestrator code must be updated to provide the necessary hooks. This involves two key additions:

1. **A Look-Ahead Prefetch Ring:** The CPU must maintain a ring buffer of upcoming expert selections. This ring buffer is allocated in pinned host memory, and its address is passed to the GPU at kernel launch time. The GPU kernel uses this to know which expert to prefetch, and the CPU uses it to keep the GPU's pipeline full.
2. **A Semaphore Register:** The CPU must allocate a single, 32-bit memory location in pinned host memory and map it to the GPU's `__constant__` memory space. Before launching the next expert weight loading kernel, the CPU will increment this

semaphore, which will immediately wake up the GPU wavefront waiting on `s_sem_wait`.

The CPU-side initialization and signaling would look like this:

```
// On the CPU, during initialization
uint32_t* d_semaphore;
hipHostMalloc(&d_semaphore, sizeof(uint32_t));
hipHostRegister(d_semaphore, sizeof(uint32_t), 0);
hipMemcpy(d_semaphore, &host_semaphore, sizeof(uint32_t), hipMemcpyHostToDevice);

// Before launching the kernel for expert X
host_semaphore = X; // Set the identifier
hipMemcpy(d_semaphore, &host_semaphore, sizeof(uint32_t), hipMemcpyHostToDevice);
// Launch kernel that waits on g_expert_semaphore
```

This creates a clean, lock-free communication channel. The GPU kernel spends minimal cycles in a low-power state, and the CPU can signal readiness with a single, fast memory write. This eliminates the `WAVE_DEP_WAIT` stalls that occur when the GPU is dependent on a CPU signal, further reducing the overall `WAVE_ISSUE_WAIT` metric and driving the system closer to the theoretical maximum throughput of the PCIe bus [[1]].



Validation and Risk Assessment Framework

The success of these micro-architectural optimizations is not measured in abstract claims but in concrete, measurable improvements to hardware counters and benchmark metrics. A rigorous validation framework is therefore essential to confirm that the intended effects are achieved and to identify any unintended consequences.

Key Performance Metrics and Validation Methods

The validation process must be multi-layered, focusing on both the hardware-level impact and the end-to-end application performance.

1. **Reduction in `WAVE_ISSUE_WAIT` Cycles:** This is the primary indicator of success for the SDMA pipelining and hardware signaling optimizations. A successful implementation should reduce the `WAVE_ISSUE_WAIT` metric by at least 15% compared to the baseline. This reduction signifies that the scheduler is no longer

starved for instructions due to data dependencies and that the GPU is able to find sufficient independent work to keep its pipelines full [[1]]. The validation method is to capture telemetry using `rocprofv3` with a specific counter file (`counters_sdma.txt`) that focuses on `WAVE_ISSUE_WAIT`, `WAVE_DEP_WAIT`, and `MeanOccupancyPerCU`. The test must be run on the `Qwen3_35BMTPIQ4.gguf` model with the `-ngl 99 -ncmoe 16` flags to ensure the MoE routing logic is fully engaged [[1]].

2. **Increase in MemUnitBusy Utilization:** For the zero-copy GTT access, the goal is to push the `MemUnitBusy` metric as close to 100% as possible. The baseline of 85.29% indicates that the memory unit is underutilized, suggesting that the GPU is spending too much time waiting for commands or coherency protocols instead of performing active data transfers [[1]]. A successful implementation of `GLOBAL_LOAD_DWORD` with `SLC=1` should demonstrate a measurable increase in this metric, reflecting a more efficient use of the available PCIe bandwidth. This is validated by comparing the `MemUnitBusy` value from the `rocprofv3` output before and after the patch is applied [[1]].
3. **End-to-End Token Throughput (tg128):** The ultimate measure of success is the user-facing performance metric: tokens generated per second. The validation gate for the entire suite is a minimum increase of 5% in `tg128` throughput compared to the `v0.3.0-stable` baseline, with a strict variance cap of $\leq \pm 1.5$ t/s across 5 runs [[1]]. This ensures that the optimization delivers real-world benefits and does not merely shift bottlenecks elsewhere. The benchmark must be executed with identical configurations, including the same ROCm version, model, quantization, and context length, to ensure a fair comparison.

Primary Risks and Mitigation Strategies

Every hardware-level optimization carries risk, and the most significant risks are associated with violating the foundational assumptions of the design.

1. **Incorrect `s_sendmsg` Configuration:** The most severe risk is misconfiguring the `s_sendmsg` instruction, which could lead to SDMA queue corruption, data corruption, or a complete system hang. The mitigation is a thorough review of the SDMA descriptor format and the `s_sendmsg` operand encoding as specified in the RDNA2 ISA manual [[1]]. The implementation must be tested with a small, isolated kernel that triggers a single, well-understood transfer before being integrated into the main inference path.

2. **Using SLC=1 on Unpinned Memory:** This is the most critical risk. If the SLC=1 flag is used with memory that has not been properly pinned with `hipHostMalloc`, the hardware will attempt to perform a direct, uncached read from the virtual address. This will inevitably trigger an OS page fault, which must be handled by the operating system. The resulting latency is measured in milliseconds, not nanoseconds, and would completely obliterate any performance gain [[1]]. The mitigation is twofold: first, the implementation must include a mandatory runtime check at startup that verifies all required host memory allocations are indeed pinned. Second, the documentation and build process must enforce the use of pinned memory as a non-negotiable requirement, clearly stating that the optimization suite is incompatible with unpinned memory [[1]].
3. **Race Conditions in Hardware Signaling:** While `s_sem_wait` is designed to be atomic, improper initialization of the semaphore or incorrect sequencing of CPU writes and GPU waits can lead to deadlocks or missed signals. The mitigation is to establish a strict initialization protocol. The CPU must initialize the semaphore to a value of 0, and the GPU kernel must be launched only after the CPU has confirmed the semaphore is ready. This creates a clear, unidirectional flow of control that eliminates ambiguity.

By rigorously validating against these metrics and proactively mitigating these risks, the optimization suite can be deployed with confidence, transforming the PCIe bus from a performance bottleneck into a seamlessly integrated component of the high-throughput inference pipeline.



End-to-End Implementation Blueprint for the `ggml-hip` Backend

The preceding analysis has dissected the RDNA2 ISA to identify the most potent hardware primitives for hiding PCIe latency. This section synthesizes those primitives into a cohesive, production-ready implementation plan for the `ggml-hip` backend. It moves beyond theory to provide the precise, actionable steps required to integrate these optimizations, specifying the exact files, functions, and instruction sequences that must be modified.

Kernel-Level Modifications: Integrating the ISA Primitives

The integration of the three pillars must be implemented in a way that respects the existing codebase structure while injecting the new hardware capabilities at the most strategic points.

- 1. SDMA Pipeline Integration in `mmvq.cu`:** The `mul_mat_vec_q` kernel is the central hub of the inference pipeline. Its modification is the highest priority. The kernel must be augmented to include a prefetching stage at the end of its main loop. This stage will be implemented as a separate, conditional code block, guarded by a `#ifdef RDNA2_MOE_SDMA_V1` preprocessor directive to ensure reversibility [[1]]. The block will use the `s_sendmsg` instruction to write a descriptor to the SDMA queue, followed by `s_dcache_wb` and `s_waitcnt` to ensure safety and proper ordering [[1]]. The descriptor structure itself must be defined in a new header file, `ggml/src/ggml-hip/sdma_descriptor.h`, which will contain the exact bitfield definitions for `SRC_ADDR_LO`, `DST_ADDR_LO`, `BYTE_COUNT`, and `CONTROL` as specified in the RDNA2 ISA manual [[1]]. This separation of concerns makes the descriptor definition reusable and easily auditable.
- 2. Zero-Copy GTT Access in `ggml-hip-get-rows.cuh`:** The `get_rows` functionality is the natural home for the zero-copy access pattern. A new, overloaded function, `ggml_get_rows_gtt`, should be added. This function will be compiled only when `RDNA2_MOE_GTTCOPY_V1` is defined [[1]]. Its implementation will consist of an inline assembly block that uses the `GLOBAL_LOAD_DWORD` instruction with the `slc` modifier to load a single gating network weight directly into a VGPR [[1]]. The function signature will be `float ggml_get_rows_gtt(const float* gating_weights, int idx)`, mirroring the existing `get_rows` interface for seamless integration.
- 3. Wavefront Synchronization in `convert.cu`:** The `convert.cu` file contains the kernels responsible for converting and preparing data for the GPU. The expert weight loading kernel, which is responsible for moving weights from the staging buffer into the final compute buffers, is the logical place to implement the `s_sem_wait` instruction [[1]]. At the very beginning of this kernel, a new line will be added: `__builtin_amdgcn_s_sem_wait(&g_expert_semaphore, 1);`. This line will be guarded by the `#ifdef RDNA2_MOE_SYNC_V1` directive [[1]]. The `g_expert_semaphore` is declared as a `__constant__` variable at the top of the file, ensuring it resides in the GPU's constant memory space, which is accessible for hardware signaling [[1]].

CPU-to-GPU Signaling Contract: Defining the Interface

The GPU optimizations are useless without a corresponding change in the CPU orchestrator. The `ggml` library's main CPU code must be updated to provide the necessary infrastructure. This is not a matter of modifying dozens of files, but of adding a few critical, well-defined interfaces.

1. **Look-Ahead Prefetch Ring Buffer:** A new, lightweight ring buffer class, `ggml_moe_prefetch_ring`, must be added to `ggml/src/ggml.c`. This class will be instantiated by the MoE orchestrator and managed as part of the model's state. It will be allocated in pinned host memory using `hipHostMalloc` and will contain a fixed number of entries, each holding the index of an upcoming expert to be prefetched [[1]]. The GPU kernel will read from this ring to know which expert to load next, effectively turning the CPU's routing decision into a pre-emptive, hardware-managed prefetch.
2. **Semaphore Register Initialization:** The CPU orchestrator must allocate and initialize the semaphore that the GPU kernel will wait on. This is done in `ggml/src/ggml-hip/ggml-hip.c`, within the `ggml_backend_hip_init` function. Here, a 32-bit integer is allocated on the host, registered as pinned memory, and its device address is passed to the GPU using `hipMemcpy` to initialize the `g_expert_semaphore` constant [[1]]. This establishes the shared memory location for communication.
3. **Signal Protocol:** The final piece of the contract is the signal protocol. Within the MoE routing logic, after the CPU has selected the next expert and prepared its weight data in the staging buffer, it must signal the GPU. This is a single line of code: `hipMemcpy(g_expert_semaphore_dev_ptr, &next_expert_id, sizeof(uint32_t), hipMemcpyHostToDevice);`. This simple, fast memory copy is the trigger that wakes up the GPU wavefront, completing the asynchronous, lock-free communication loop [[1]].

This contract is deliberately minimal and explicit. It does not require a rewrite of the MoE logic; it only adds the necessary scaffolding to enable the GPU to operate autonomously. The CPU remains in control of the high-level routing, while the GPU takes full responsibility for managing the low-level data movement and synchronization.

Strategic Conclusion: Transforming the PCIe Bus into a Seamless Extension of the GPU

The journey through the RDNA2 ISA manual reveals a profound truth about modern GPU architecture: the boundaries between the "compute" and "I/O" domains are increasingly artificial. The RX 6800 XT is not just a collection of SIMD cores and a memory controller; it is a sophisticated, integrated system-on-chip with dedicated, programmable engines for DMA, a rich, configurable memory hierarchy, and hardware primitives for inter-processor communication. The path to unlocking its full potential for MoE inference lies not in attempting to make the GPU faster, but in rethinking the entire data flow as a holistic, hardware-accelerated process.

The three-pillar strategy outlined in this report is the direct result of this systems-level perspective. SDMA pipelining treats the PCIe bus as an extension of the GPU's own memory subsystem, allowing the GPU to manage its own data pipeline with autonomous, hardware-triggered transfers. Zero-copy GTT access acknowledges that for certain workloads, the fastest memory is not the one closest to the GPU, but the one that is most efficiently accessed, even if it resides in system RAM. By using `GLOBAL_LOAD_DWORD` with `SLC=1`, we bypass the cache hierarchy not as a failure, but as a deliberate strategy to preserve the integrity of the critical KV cache and to streamline the data path [[1]].

Finally, hardware signaling with `s_sem_wait` dissolves the artificial wall between the CPU and GPU. Instead of a client-server relationship where the GPU waits for a command from the CPU, it becomes a peer-to-peer relationship where the GPU is an active participant in the orchestration, capable of expressing its readiness and waiting for the CPU's signal with near-zero overhead. This transforms the "PCIe bottleneck" from a rigid, immutable constraint into a flexible, tunable resource.

The outcome of this strategy is a fundamental shift in the performance profile. The `WAVE_ISSUE_WAIT` metric, once the dominant source of latency, is reduced, allowing the scheduler to keep the VALU units busy. The `MemUnitBusy` metric increases, indicating that the memory unit is being driven harder and more efficiently. And ultimately, the `tg128` token throughput rises, delivering tangible, user-facing benefits. This is the essence of `Code in Full Review`: the hardware, through its telemetry counters, tells us what is working and what is not. The RDNA2 ISA manual is the map, and this report is the guidebook for using that map to navigate the complex terrain of MoE inference on the RX 6800 XT. By treating the entire system as a single, unified entity, we can finally turn the promise of massive MoE models into a reality that runs at maximum efficiency on our hardware [[1]].



References

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